



THE SCHOOL OF BUSINESS & ECONOMICS

FixCity: Smart Urban Maintenance & Service Platform



Abstract

Rapid urbanization in Egypt has led to increasing challenges in maintaining street infrastructure, often resulting in prolonged repair times for issues like potholes, broken lighting, and waste accumulation. Fixcity is a cross-platform mobile application and web dashboard designed to bridge the communication gap between citizens and municipal authorities. By leveraging geolocation technology and real-time database synchronization, the platform allows citizens to document and report non-emergency issues instantly. The system creates a transparent workflow where reports are tracked via unique codes, and authorities can manage tasks through a centralized admin panel. This project demonstrates how modern, open-source technologies (Flutter & Supabase) can be utilized to foster civic engagement and improve urban living standards.

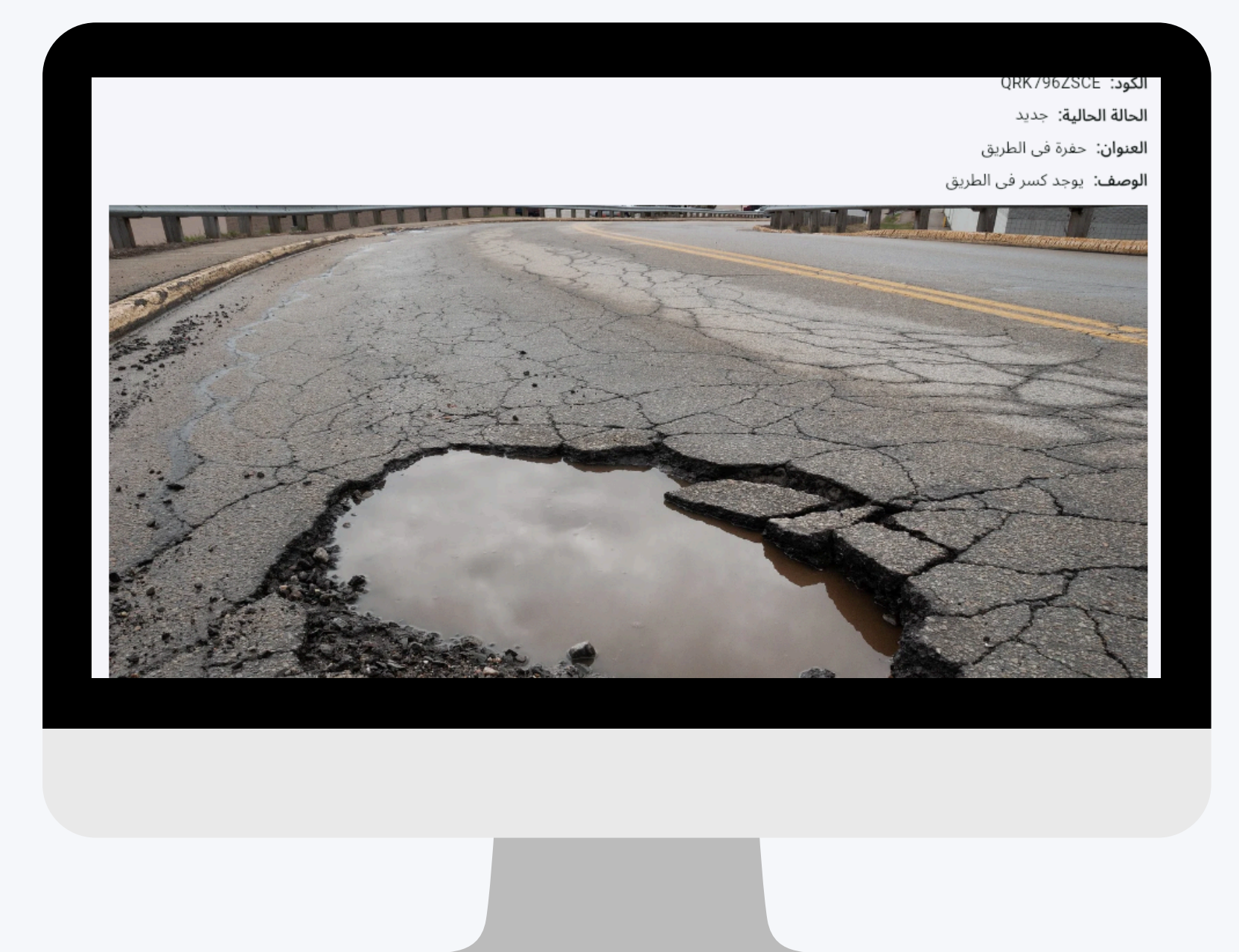
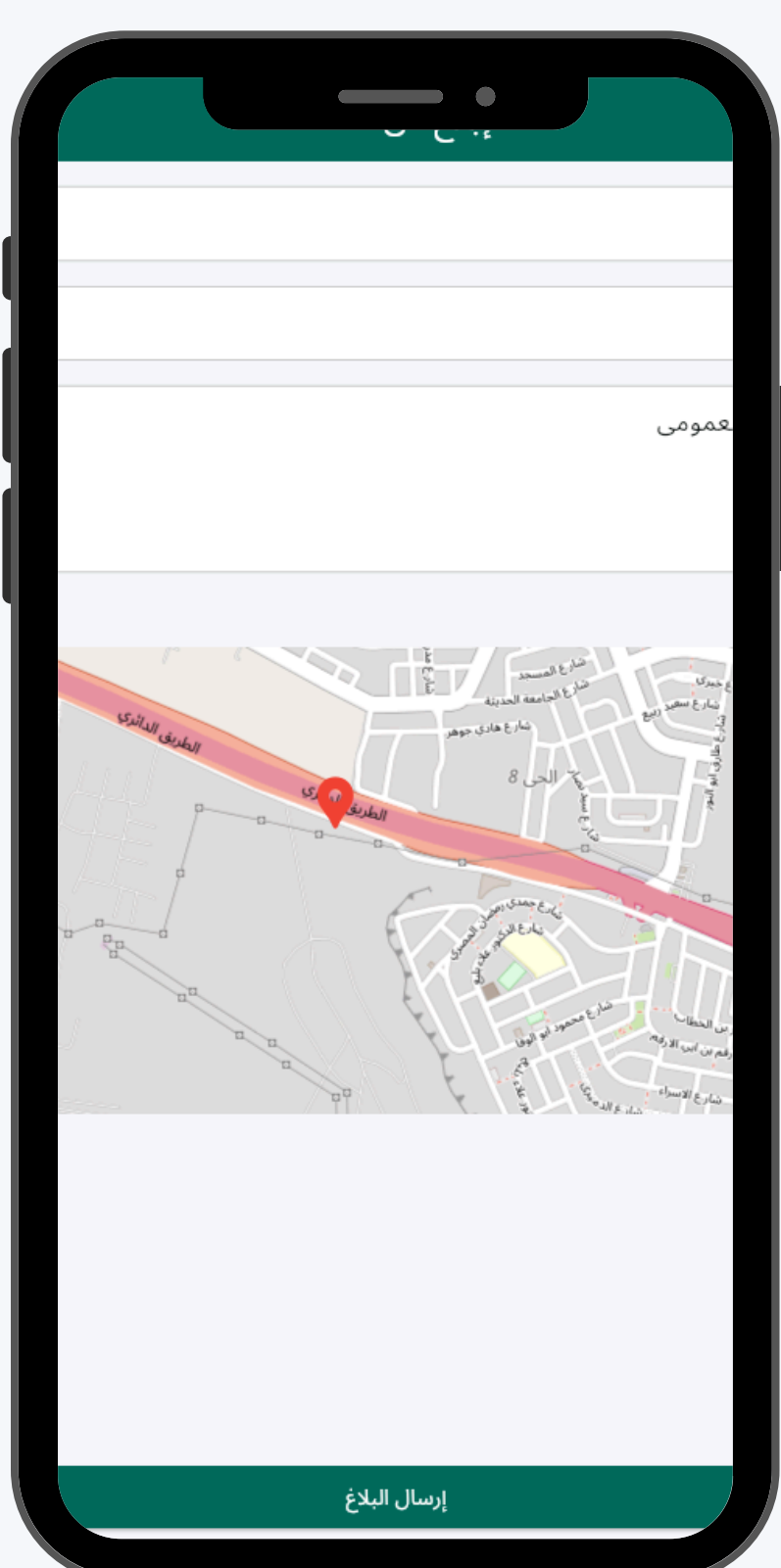
Research Problem

1. Lack of Precision: Verbal descriptions often fail to pinpoint exact locations.
2. Opacity: Citizens rarely receive updates on their complaints, leading to distrust.
3. Data Fragmentation: Authorities lack a unified digital map of infrastructure issues. The research addresses the need for a unified, digital-first reporting ecosystem that ensures accountability and accessibility for all demographic groups.

Methodology

- Frontend: Flutter was selected to deploy a single codebase across Android, iOS, and Web platforms.
- Backend: Supabase was utilized to leverage PostgreSQL for relational data integrity and Row Level Security (RLS) for data protection.
- Geolocation: Integrated OpenStreetMap (via Leaflet) to provide accurate coordinate tracking and visualization.
- Data Flow: A reactive data model is used where status updates on the dashboard are instantaneously reflected on the user app without manual refreshing.

Prototype



Recommendations

1. AI Integration: Implementing computer vision to automatically categorize problems from the uploads.
2. Gamification: Introducing a points system to reward users for verified reports, encouraging participation.
3. Offline Mode: Developing a local caching mechanism to allow report drafting in areas with poor network.
4. Government API Integration: Connecting directly with municipal work-order systems for automated dispatching.

Author

Ahmed Wael Ali
Shehab El-Amir
Moataz Ahmed Shaker
Mohamed Salaheldin
George Armia
Mohamed Ahmed
Abdel-Salam

Submitted to:
Prof. Dr. Wafaa el-Shereif
Prof. Dr. Mohamed
Dr. Omnia Ahmed